

Opinion Dynamics with Multiple Adversaries

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Abstract

Opinion dynamics models how the publicly expressed opinions of users in a social network coevolve according to their neighbors as well as their own intrinsic opinion. Motivated by the real-world manipulation of social networks during the 2016 US elections and the 2019 Hong Kong protests, a growing body of work models the effects of a strategic actor who interferes with the network to induce disagreement or polarization. We lift the assumption of a single strategic actor by introducing a model in which any subset of network users can manipulate network outcomes. They do so by acting according to a fictitious intrinsic opinion. Strategic actors can have conflicting goals, and push competing narratives. We characterize the Nash Equilibrium of the resulting meta-game played by the strategic actors. Experiments on real-world social network datasets from Twitter, Reddit, and Political Blogs show that strategic agents can significantly increase polarization and disagreement, as well as increase the “cost” of the equilibrium. To this end, we give worst-case upper bounds on the Price of Misreporting (analogous to the Price of Anarchy). Finally, we give efficient learning algorithms for the platform to (i) detect whether strategic manipulation has occurred, and (ii) learn who the strategic actors are. Our algorithms are accurate on the same real-world datasets, suggesting how platforms can take steps to mitigate the effects of strategic behavior.

CCS Concepts

• **Information systems** → **World Wide Web; Social networks; Electronic commerce.**

Keywords

platform design; misreporting; social networks

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1 Introduction

Over the past decade, social media platforms have grown dramatically in both scale and societal influence, enabling billions of users to share information and opinions instantaneously. Online social networks now play a central role in how individuals consume news, form political views, and engage with issues related to health, products, and public life [4, 5, 57, 61].

A growing body of evidence suggests, however, that these platforms can exacerbate polarization by shaping and amplifying patterns of social interaction [15, 26, 44, 60]. One prominent explanation is the *filter-bubble* theory [48], whereby personalized algorithms selectively expose users to content aligned with their past behavior and beliefs. This form of “invisible algorithmic editing” limits exposure to diverse viewpoints, reinforcing existing opinions and potentially undermining democratic discourse by reducing engagement with challenging or unfamiliar ideas.

Beyond endogenous polarization, social networks are increasingly subject to strategic manipulation by malicious actors seeking to induce discord. Notable examples include the alleged interference of the Russian Internet Research Agency in the 2016 U.S. presidential election [43], coordinated attempts to disrupt protest organizations in Hong Kong [59], and large-scale scams exploiting political and conspiratorial content to maximize engagement and profit [12]. These interventions leverage the economics of attention on digital platforms, where inflammatory content is rewarded and algorithmically amplified, making social networks particularly vulnerable to strategic misuse.

To study the evolution of beliefs in such settings, researchers across computer science, sociology, and statistics have relied on models of *opinion dynamics*, in which agents’ expressed opinions coevolve over a weighted network by combining intrinsic beliefs with social influence from neighbors [25]. While this framework captures the interplay between individual predispositions and network effects, existing work largely focuses on scenarios involving a single adversarial or coordinating actor acting on the network to induce polarization or disagreement [15, 19, 26, 44, 50, 51, 60].

In this work, we lift the assumption of requiring a single actor (such as the platform) to act as an adversary to induce polarization or disagreement and consider the case of several decentralized actors. It is known that empirically, a very small percentage (25%) of the users in a network need to disagree to sway consensus [13]. Moreover, real-world social networks involve *multiple* malicious actors, who use different levels of manipulation and hate speech based on their individual goals [12]. In this paper, we attempt to

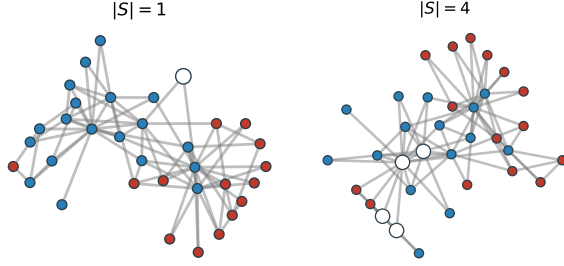


Figure 1: Visualization of the strategic equilibrium (z') on the Karate Club Graph for two different choices of S . The truthful intrinsic opinions have been taken to be $s = u_2$ where u_2 is the Fiedler eigenvector of G . The white nodes correspond to the nodes in S . For the other nodes, the nodes colored in blue (resp. red) correspond to nodes whose public opinion z'_i increased (resp. decreased), i.e., $(z'_i - z_i)/z_i \geq 0$ (resp. $(z'_i - z_i)/z_i < 0$) after s' was chosen.

provide a theoretical basis for this phenomenon: specifically, in our setting, we assume that there is a set $S \subseteq V$ of strategic agents whose goal is to report false intrinsic opinions (s') that are different from their true intrinsic opinions ($s \neq s'$). Their goal is to influence others while not deviating much from their neighbors; namely, they want to reach an equilibrium where their neighbors agree with them.

For instance, assume a social network where a set of S of political actors want the network to believe that their stance on a topic (e.g., abortion, elections, drug legalization, etc.) is the best. They achieve this by adversarially reporting different intrinsic opinions. This ensures that their influence is both persuasive and credible within the local network context. Such adversarial behavior can result in significantly different (cf. Figure 1) and highly polarized equilibria, where the strategic agents' opinions appear dominant despite not reflecting the actual intrinsic views of the majority.

Our work investigates the conditions under which these strategic manipulations are successful, the extent of their impact on network-wide opinion dynamics, and how platforms can learn from observing these manipulated equilibria to mitigate such impacts.

In this paper, we ask the following research question (RQ):

(RQ) What if a set of strategic actors with possibly conflicting goals tries to manipulate the consensus by strategically reporting beliefs different than their true beliefs?

We rely on the Friedkin-Johnsen (FJ) model [25], where the opinions of agents coevolve via the help of a weighted undirected network $G = (V = [n], E, w)$ with non-negative weights. According to the FJ model, the agents possess intrinsic opinions s and express opinions z , which they update via the following rule for each agent i :

$$z_i(t+1) = \frac{\alpha_i s_i + (1 - \alpha_i) \sum_{j \sim i} w_{ij} z_j(t)}{1 + \sum_{j \sim i} w_{ij}}. \quad (1)$$

where $\alpha_i \in (0, 1)$ is i 's susceptibility to persuasion [1]. We additionally define $\tilde{\alpha}_i = \alpha_i / (1 - \alpha_i)$ to be the normalized susceptibility parameter corresponding to i . This update rule corresponds to the

best-response dynamics arising from minimizing the quadratic cost function for each $i \in [1, 10]$:

$$c_i(z_i, z_{-i}) = (1 - \alpha_i) \sum_{j \sim i} w_{ij} (z_i - z_j)^2 + \alpha_i (z_i - s_i)^2. \quad (2)$$

The Pure Strategy Nash Equilibrium (PSNE) can be written as $z = ((I - A)L + A)^{-1} A s = B s$ where L is the Laplacian of graph G , $A = \text{diag}(\alpha_1, \dots, \alpha_n)$ is the diagonal matrix of susceptibilities. When an external *single actor* aims to induce disagreement or polarization – see, e.g., Gaitonde et al. [26], Musco et al. [44], Racz and Rigobon [49] – the adversary is coined with optimizing the objective function

$$\sum_{i \in [n]} c_i(z_i, z_{-i}) = s^T ((I - A)L + A)^{-1} A f(L) ((I - A)L + A)^{-1} A s,$$

where $f(L)$ is a function of the Laplacian of G , either with optimizing towards s [26], or the graph itself [44, 49].

Usually, as we also discussed earlier, many adverse actions on social networks come from *several independent strategic adversaries* who try to manipulate the network by infiltrating intrinsic opinions s'_i , which are different from their true stances s_i but are simultaneously close to s_i . Unlike previous works, these “adversaries” can have conflicting goals.

Concretely, the true opinions of the agents are $s_1, \dots, s_n \in \mathbb{R}$, and there is a set S of deviating agents who report $\{s'_i\}_{i \in S}$. While an agent $i \in S$ has a true intrinsic opinion s_i , she may report some $s'_i \neq s_i$ for the purpose of manipulating the dynamics of Equation (1). Specifically, her goal is to minimize her unique cost function, given by Equation (2), at the resulting consensus $z' = ((I - A)L + A)^{-1} A s'$ where s' is the vector which has entries s_i for all $i \notin S$ and s'_i for all $i \in S$. The local optimization of agent i becomes:

$$\min_{s'_i \in \mathbb{R}} c_i(z' = ((I - A)L + A)^{-1} A s'). \quad (3)$$

Note that while Equation (3) depends on the choices of all $j \in S$, agent i can only choose the i^{th} entry of s' . Our contributions are as follows.

Characterizing Nash Equilibria with Multiple Adversaries. We give the Nash equilibrium of the game defined by Equation (3), and show that all Nash-optimal strategies are pure. The Pure Strategy Nash Equilibrium (PSNE) that is given by solving a constrained linear system. Given the PSNE of the game, we characterize the actors who can have the most influence in strategically manipulating the network.

Real-World Experiments to Understand Properties of Equilibria. We apply our framework to real-world social network data from Twitter and Reddit [19], and data from the Political Blogs (Polblogs) dataset [2]. We find that the influence of strategic agents can be rather significant as they can significantly increase polarization and disagreement, as well as increase the overall “cost” of the consensus.

Analysis of Equilibrium Outcomes Under Different Sets of Strategic Actors. Various metrics for network polarization and disagreement are sensitive to the choice of *who* acts strategically, in nontrivial ways. For example, adding more strategic agents can sometimes *decrease* the Disagreement Ratio at equilibrium (Figure 5), due to

counterbalancing effects. To address the effects of manipulation, we give worst-case upper bounds on the *Price of Misreporting* (PoM), which is analogous to well-studied Price of Anarchy bounds (see, for example, Bhawalkar et al. [9], Roughgarden and Schoppmann [55]), and suggest ways that the platform can be used to mitigate the effect of strategic behavior on their network.

Learning Algorithms for the Platform. We give an efficient algorithm for the platform to detect if manipulation has occurred (Algorithm 1), based on a hypothesis test with the publicly reported opinions z' . Next, we give an algorithm to infer *who* manipulated the network (the set of strategic agents S) from z' , as long as the size of S is sufficiently small. Our algorithm is inspired by the robust regression algorithm of Bhatia et al. [7], and is practical for real-world networks. It (i) requires the platform to have access to node embeddings X which have been shown computable even in billion-scale networks such as Twitter [22], and (ii) can be computed in time $(n + m)^{O(1)}$, where n is the number of nodes and m is the number of edges of the network. Our algorithms have high accuracy on real-world datasets from Twitter, Reddit, and Polblogs.

1.1 Preliminaries and Notations

The set $[n]$ denotes $\{1, \dots, n\}$. $\|x\|_p$ denotes the ℓ_p -norm of vector x , whose i -th entry is denoted by x_i . $X \succeq 0$ denotes that the matrix X is positive semi-definite and $\|X\|_2$ corresponds to the spectral norm of X . The Laplacian of the graph G is denoted by $L = D - W$ where W is the weight matrix of the graph, which has entries $w_{ij} \geq 0$, and D is the diagonal degree matrix with diagonal entries $D_{ii} = \sum_j w_{ij}$. The Laplacian has eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. For any undirected and connected graph G , L is symmetric and PSD, so we can write the eigendecomposition of L as:

$$L = \sum_{i \in [n]} \lambda_i u_i u_i^T \succeq 0, \quad (4)$$

where $u_1, \dots, u_n$ are orthonormal eigenvectors. Moreover, $u_1 = (1/\sqrt{n})\mathbf{1}$, where $\mathbf{1}$ is the column vector of all 1s. U denotes the matrix which has the eigenvectors of L as columns; i.e., such that $L = U^T \Lambda U$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ is the diagonal matrix of L 's eigenvalues. L_i denotes the i -restricted Laplacian which corresponds to the Laplacian of the graph with all edges that are non-adjacent to i being removed, and, similarly, $L_{\{u,v\}}$ corresponds to the Laplacian of an edge $\{u, v\}$. Note that $L_i = \sum_{j \sim i} L_{\{i,j\}}$. For a function $f(L)$ of the Laplacian we write $f(L) = U^T f(\Lambda) U$ where $f(\Lambda) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n))$. For brevity, regarding the equilibrium z of the FJ model, we write $B = ((I - A)L + A)^{-1}A$, such that $z = Bs$ and $z' = Bs'$. Finally $e_1, \dots, e_n \in \mathbb{R}^n$ denote the canonical basis.

We define the total cost of an equilibrium z to be

$$C(z) = \sum_{i \in [n]} c_i(z). \quad (5)$$

We define the platform-wide metrics to be

$$\text{Polarization} \quad \mathcal{P}(z) = \sum_{i \in [n]} (z_i - \bar{z})^2, \bar{z} = \frac{1}{n} \sum_{i \in [n]} z_i, \quad (6)$$

$$\text{Disagreement} \quad \mathcal{D}(z) = \sum_{i,j \in [n]} w_{ij} (z_i - z_j)^2 = z^T L z. \quad (7)$$

Finally, we define the *“Price of Misreporting”* (PoM), which is analogous to the Price of Anarchy [54]. The PoM is the ratio of the cost $C(z')$ when the agents are deviating, and the cost $C(z)$ when the agents are reporting truthfully, i.e.,

$$\text{PoM} := \frac{C(z')}{C(z)}. \quad (8)$$

Unlike the Price of Anarchy (PoA), the equilibrium z in the denominator of Eq. (8) is the Nash equilibrium for the Friedkin-Johnson dynamics without manipulation. In the PoA, the denominator would be $C(z^*)$, where z^* is a socially optimal equilibrium [10]. Since we study strategic manipulations as a meta-game with respect to the base game of FJ dynamics, it is more relevant for us to compare z' with z than with z^* .

1.2 Related Work

Opinion Dynamics. Opinion dynamics are well-studied in computer science and economics, as well as sociology, political science, and related fields. There have been many models proposed for opinion dynamics, such as with network interactions as we study in this paper (FJ model) [11, 25], bounded confidence dynamics (Hegselman-Krauss Model) [33], coevolutionary dynamics [9] as well as many variants of them; see, for example Abebe et al. [1], Fotakis et al. [23, 24], Hązła et al. [32], Ristache et al. [51]. The work of [10] shows bounds on the Price of Anarchy (PoA) between the PSNE and the welfare-optimal solution for the FJ model, and the subsequent work of [9] shows PoA bounds for the coevolutionary dynamics. Additionally, the opinion dynamics have been modeled by the control community; see, for example, [8, 14, 21, 46].

As in these works, we treat the FJ model as a basis. However, our work is significantly different as it studies a framework where any subset $S \subseteq [n]$ of strategic agents can *deviate* from their truthful intrinsic opinions, as opposed to studying the evolution of the expressed opinions and their PSNE in the FJ model. In our model, each strategic agent $i \in S$ can only choose a single entry s'_i of the overall deviation s' , but pays a cost based on the resulting equilibrium ($z' = Bs'$), which depends on the choices of other members of S .

Disagreement and Polarization in Social Networks. Motivated by real-world manipulation of social networks in, e.g., the 2016 US election, a recent line of work studies polarization and strategic behavior in opinion dynamics [16, 26, 27, 51, 52, 60]. Chen and Rácz [16] consider a model in which an adversary can control $k \leq n$ nodes' internal opinions and seeks to maximize polarization at equilibrium. Similarly, Gaitonde et al. [26] considers a single adversary who can modify intrinsic opinions s belonging to an ℓ_2 -ball. More recent work also studies modification of agents' susceptibility parameters α_i to alter the median opinion at equilibrium [52]. Our work shares a similar perspective to these: non-strategic agents follow the base-game (in our case, Friedkin-Johnson dynamics Equation (1)), while strategic agents treat one or more variables in the base game as a *strategic choice* that they may vary to manipulate game outcomes (in our case, the misreported intrinsic opinions $\{s'_i\}_{i \in S}$). This perspective emphasizes the dynamics and equilibria of the resulting meta-game, which is played by the strategic agents

against the backdrop of the base-game dynamics. Unlike these previous works, however, in our setting any subset $S \subseteq [n]$ can be strategic, and these “adversaries” can have conflicting goals.

Manipulation of Dynamic Games. Opinion dynamics are a widely studied instance of a network game, which is a game played by nodes in a network with payoffs depending on the actions of their neighbors [38, 58]. In addition to the manipulation of opinion dynamics, researchers have studied strategic manipulation of financial network formation [35]. In the non-network setting, researchers have studied the manipulation of recommendation systems from a game-theoretic perspective [6], as well as security games [47], repeated auctions [40] and Fisher markets with linear utilities [39].

Learning from Strategic Data. We develop learning algorithms which observe the (possibly manipulated) equilibrium $\mathbf{z}'$ to detect if manipulation occurred, and if so who was responsible. The former problem relates to anomaly detection in networks. Chen and Tsourakakis [17] develop a hypothesis test to detect such fraud in financial transaction networks, by testing if certain subgraphs deviate from Benford’s Law. Similarly, Agarwal et al. [3] propose a framework based on a χ^2 -statistic to perform graph similarity search.

The problem of recovering the set of deviators relates to the broader literature of learning from observations of network games. Most works give learning algorithms for games *without* manipulation [20, 28, 34, 36, 41, 53]. But our data $\mathbf{z}'$ can be a manipulated equilibrium, which is a *strategic source* of data [62]. Learning algorithms for strategic sources are known for certain settings such as linear classifiers with small-deviation assumptions [18], or binary classifiers in a linear reward model [31]. When agents can modify their features to fool a known algorithm, even strategy-robust classifiers such as [30] can be inaccurate [29]. Since agents can deviate arbitrarily in our model, we use a robust regression method with guarantees against *adversarial* corruptions [7], similar to the learning algorithms in [37, 56]. The work of Jalan and Chakrabarti [35] studies learning from financial networks with strategic manipulations, which is in a similar spirit to our work but differs significantly in the application domain and context.

Real-world Datasets. To support our results, we use data grounded in practice, which have also been used in previous studies to study polarization and disagreement (cf. Adamic and Glance [2], Chitra and Musco [19], Musco et al. [45], Wang and Kleinberg [60]). Specifically, we use Twitter, Reddit, and Political blog networks. Both the Twitter and Reddit datasets are due to Chitra and Musco [19]. The vectors $\mathbf{s}$ of initial opinions for both are obtained via sentiment analysis and also follow the post-processing of Wang and Kleinberg [60].

(1) *Twitter dataset* ($n = 548, m = 3638$). These data correspond to debate over the Delhi legislative assembly elections of 2013. Nodes are Twitter users, and edges refer to user interactions.

(2) *Reddit dataset* ($n = 556, m = 8969$). These data correspond to political discussion on the *r/politics* subreddit. Nodes are who posted in the *r/politics* subreddit, and there is an edge between two users i, j if two subreddits (other than *r/politics*) exist that both i, j posted on during the given time period.

(3) *Political Blogs (Polblogs) dataset* ($n = 1490, m = 16178$). These data, due to Adamic and Glance [2], contain opinions from political blogs (liberal and conservative). Edges between blogs were automatically extracted from a crawl of the front page of the blog. Each blog is either liberal – where we assign a value $s_i = -1$ – or conservative – where we assign $s_i = +1$.

2 Strategic Opinion Formation

The opinion formation game has two phases. First, strategic agents privately choose a strategic intrinsic opinion according to Equation (3). Second, agents exchange opinions and reach consensus as if they were in the Friedkin-Johnson dynamics, except the strategic opinions are used in place of the true intrinsic opinions.

- (1) *Strategy Phase.* Each strategic agent $i \in S$ independently and privately chooses a fictitious strategic opinion $\mathbf{s}'_i \in \mathbb{R}$. For honest agents ($i \notin S$) we have $\mathbf{s}'_i = \mathbf{s}_i$.
- (2) *Opinion Formation Phase.* Reach equilibrium $\mathbf{z}' = B\mathbf{s}'$ as if $\mathbf{s}'$ were the true intrinsic opinions $\mathbf{s}$.

The network G and the true beliefs $\mathbf{s}$ determine each agent’s utility. We pose the following problem:

DEFINITION 2.1 (INTRINSIC BELIEF LYING PROBLEM.). *Let $S \subseteq [n]$ be a set of strategic agents. If agent $i \in S$ wants network members to express opinions close to $\mathbf{s}_i$, what choice of $\mathbf{s}'_i$ is optimal and minimizes the cost function of Equation (3)?*

The following theorem characterizes the Nash Equilibria of the Intrinsic Belief Lying Problem (cf. Section A for a proof).

THEOREM 2.2 (NASH EQUILIBRIUM). *Let $\mathbb{T}_i = (1 - \alpha_i)(B^T L_i B) + \alpha_i(B^T \mathbf{e}_i \mathbf{e}_i^T B) \in \mathbb{R}^{n \times n}$ and $\mathbf{y}_i = \alpha_i B_{ii} \mathbf{s}_i$. The Nash equilibria, if any exist, are given by solutions $\mathbf{s}' \in \mathbb{R}^n$ to the following constrained linear system:*

$$\begin{aligned} \forall i \in S : \mathbf{e}_i^T \mathbb{T}_i \mathbf{s}' &= \mathbf{y}_i, \\ \forall j \notin S : \mathbf{s}'_j &= \mathbf{s}_j. \end{aligned}$$

Next, we discuss some consequences of Theorem 2.2. First, we characterize $\mathbf{s}'$ as the solution to a linear system.

COROLLARY 2.3. *Let $T \in \mathbb{R}^{|S| \times n}$ have rows $\{\mathbf{e}_i^T \mathbb{T}_i\}_{i \in S}$ given by Theorem 2.2. Let $\tilde{T} \in \mathbb{R}^{|S| \times |S|}$ be the submatrix of T selecting columns belonging to S . Let $\mathbf{y} \in \mathbb{R}^{|S|}$ have entries $\mathbf{y}_i = \alpha_i B_{ii} \mathbf{s}_i$ as above. Let $\tilde{\mathbf{y}} = \mathbf{y} - \sum_{j \notin S} \mathbf{s}_j T_{ej}$. Then the set of Nash equilibria, if any exist, are given by the solutions to the unconstrained linear system*

$$\tilde{T} \mathbf{x} = \tilde{\mathbf{y}}. \quad (9)$$

The resulting opinions vector $\mathbf{s}'$ is given by $\mathbf{s}'_i = \mathbf{x}_i$ if $i \in S$ and $\mathbf{s}'_i = \mathbf{s}_i$ otherwise.

Thus, in a Nash equilibrium, every strategic agent solves their corresponding equation given by Equation (9). The explicit characterization of equilibria also implies that Nash equilibria cannot be mixed.

COROLLARY 2.4 (PURE STRATEGY NASH EQUILIBRIA). *The Nash equilibrium corresponds to solving the system of $|S|$ linear equations in the scalars $\{\mathbf{s}'_i | i \in S\}$ given by Equation (9). Also, all Nash equilibria are pure-strategy Nash equilibria.*

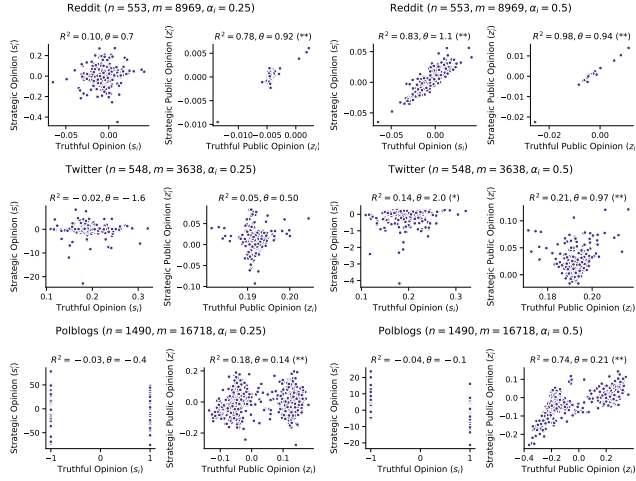


Figure 2: Plot of truthful intrinsic opinions (s) and strategic opinions (s'), and truthful public opinions (z) compared to the strategic public opinions (z') for the nodes belonging to S . S is taken to be the top-50% in terms of their eigenvector centrality. In both cases we have taken $\alpha_i \in \{0.25, 0.5\}$ for all nodes. We fit a linear regression between s' and s (resp. between z and z'). We report the effect size θ which corresponds to the slope of the linear regression and the P -value with respect to the null hypothesis ($\theta = 0$). * stands for $P < 0.001$, ** stands for $P < 0.01$ and * stands for $P < 0.05$.**

Effect of Susceptibility to Persuasion in Real-world Data. Regarding real-world data, Figure 2 shows the relationship between the truthful opinions (s and z) and the strategic ones (s' and z') for the datasets, along with the corresponding correlation coefficient R^2 , assuming that S consists of the top-50% nodes in terms of their eigenvector centrality, for susceptibility parameters set to $\alpha_i = 0.5$ (equal self-persuasion and persuasion due to others) and $\alpha_i = 0.25$ (higher persuasion due to others).

Regarding the public opinions, even though in the Reddit dataset, the strategic opinions seem to be correlated with the truthful ones ($R^2 = 0.78$ for $\alpha_i = 0.25$ and $R^2 = 0.94$ for $\alpha_i = 0.5$ respectively), in the Twitter dataset, we do not get the same result (i.e., $R^2 < 0.25$). Finally, in the Polblogs dataset, the situation is somewhere in the middle; when $\alpha_i = 0.25$ we get a low R^2 ($R^2 = 0.18$) where for $\alpha_i = 0.5$ we get a high R^2 ($R^2 = 0.74$). Additionally, in all cases except Twitter, we get that the effect is significant ($P < 0.01$).

Regarding the relationship between the intrinsic opinions, we do not detect any significant effect in most cases except Reddit with $\alpha_i = 0.5$ ($P < 0.01$) and Twitter with $\alpha_i = 0.5$ ($P < 0.05$).

Asymmetric Effects of Strategic Behavior on Liberals and Conservatives. Figure 3 analyzes the opinions of the strategic set S on the Polblogs dataset. Specifically, we find that larger changes in sentiment happen across liberal outlets compared to conservative ones. Additionally, the changes in the truthful/strategic opinions are related to the eigenvector centrality π_i as a power law, i.e., $z'_i \propto \pi_i^\theta$ ($P < 0.001$; linear regression between the log centralities $\log \pi_i$ and z'_i). The same finding holds for $|z'_i - z_i|$ and z_i .

At this point, one may wonder whether the eigenvector centrality really influences the strategic opinions z'_i for $i \in S$. Our answer

is negative. We repeat the same experiment with the Twitter and Reddit datasets, where we find no effects ($P > 0.1$; linear regression between the log centralities $\log \pi_i$ and z'_i). Due to space limitations, the corresponding figures are deferred to Section B.

Polarization and Disagreement. Figure 4 shows how the polarization, disagreement, and cost change as a function of the susceptibility parameter α_i . Except for $\alpha_i \approx 0.3$, the polarization ratio, disagreement ratio, and the price of misreporting experience a downward trend as α_i increases. This indicates that as users prioritize their own opinions more than their neighbors, they are less susceptible to strategic manipulation.

Effect of the number of deviators ($|S|$). Next, we study the effect of the number of deviators, which corresponds to $|S|$, on the changes in polarization, disagreement, and the total cost (through the price of misreporting). Figure 5 shows how the polarization and disagreement when S consists of the top-1-10% most central agents with respect to eigenvector centrality. We show that even if only 1% of agents are strategic, this can impact consensus by several orders of magnitude.

3 Price of Misreporting

In Section 2, we saw that strategic manipulation can substantially affect network outcomes via the Polarization Ratio and Disruption Ratio. We now give an upper bound for the Price of Misreporting (Eq. (8)), which is the analogue of the Price of Anarchy in our setting. The PoM measures the total cost paid by agents under the corrupted equilibrium z' , versus the total cost under the non-corrupted z . Since the cost captures an agent's deviation from her truthful intrinsic opinion as well as her deviation from the expressed opinions of her neighbors, it is a natural measure of the network's discord at equilibrium.

Theorem 3.1 shows that the PoM is small when the spectral radius of the Laplacian is small, and when agents are somewhat susceptible to their neighbors ($\alpha \rightarrow 0$). Note that the spectral radius can be replaced by a degree bound: if $d_{\max}$ is the maximum degree of the graph, then $\lambda_n \leq 2d_{\max}$. So the PoM is small if the maximum degree is small.

THEOREM 3.1. *Suppose all agents deviate ($S = [n]$) and there exists α such that $\alpha_i = \alpha$ for all i . Let $\tilde{\alpha} = \alpha/(1 - \alpha)$, and λ_n be the spectral radius of the Laplacian. Then the price of misreporting is bounded as:*

$$\text{PoM} \leq \frac{(\lambda_n + 4\tilde{\alpha})(\lambda_n + \tilde{\alpha})^2}{\tilde{\alpha}^5} = O\left(\max\left\{\frac{\lambda_n^3}{\tilde{\alpha}^5}, \frac{1}{\tilde{\alpha}^2}\right\}\right).$$

Next, we discuss how one may generalize Theorem 3.1 to the case where some agents are honest: Figure 4 shows that the PoM is *not* monotonic in $|S|$. As the number of strategic agents grows, the PoM can fall or grow, depending on the choice of S , network parameters, and so on. Therefore, we would like to give a version of Theorem 3.1 for *any* set of strategic agents $S \subset [n]$, not just the case of $S = [n]$. However, proving such a bound would require analyzing $S \times S$ principal submatrices of B, L to obtain characterizations of the cost at the corrupted equilibrium z' . In particular, we would require a *restricted invertibility* estimate to prove the analogue of Eq. (13). To our knowledge, the best such estimates [42] are too lossy when $n - |S|$ is large. We leave this question to future work.

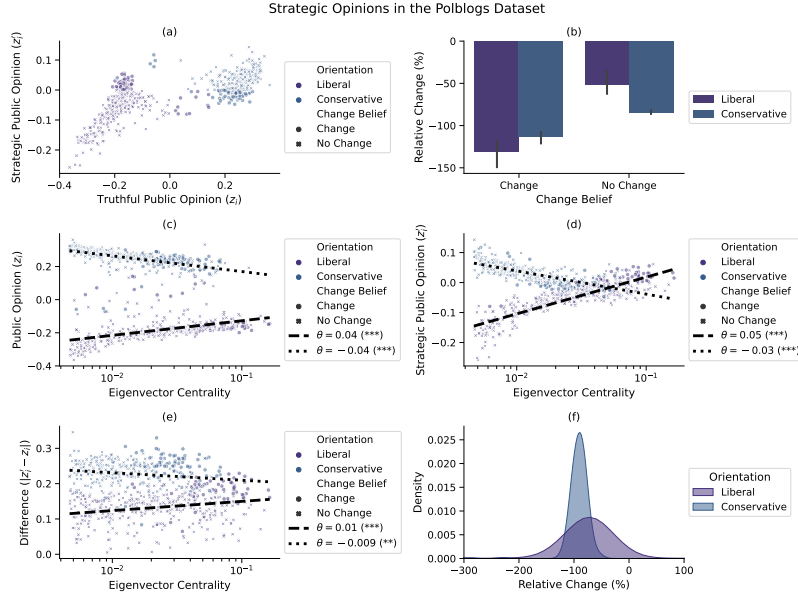


Figure 3: Strategic misreports on the Polblogs dataset, where S consists of the top 50% of agents by eigenvector centrality. Nodes are labeled as liberal ($s_i = -1$) or conservative ($s_i = +1$). A node is said to change belief if z_i and z'_i have opposite signs. In scatterplots (a), (c), (d), and (e), point shape indicates whether a belief change occurs, while color denotes the intrinsic opinion. We observe more frequent belief changes among liberal blogs than conservative ones (panel (b)). Panels (c) and (d) report truthful and strategic public opinions as functions of $\log \pi_i$, while panel (e) shows the absolute change $|z'_i - z_i|$. Regression analysis reveals significant effects of $\log \pi_i$ (*: $P < 0.001$, **: $P < 0.01$, *: $P < 0.05$; coefficients θ) on z , z' , and $|z' - z|$, consistent with power-law structure. Finally, relative belief changes are more dispersed among liberal sources than conservative ones (panel (f)).**

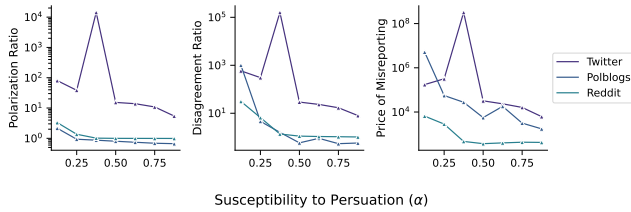


Figure 4: Polarization ratio ($\mathcal{P}(z')/\mathcal{P}(z)$), disagreement ratio ($\mathcal{D}(z')/\mathcal{D}(z)$), and price of misreporting ($C(z')/C(z)$) for the three datasets for varying susceptibility to persuasion values. We have set all susceptibilities α_i to the same value α . The Twitter dataset has the largest variation in all three ratios compared to the others. S is taken to be the top-50% nodes in terms of their eigenvector centrality.

4 Learning from Network Outcomes

To mitigate the effects of strategic behavior, a platform must understand whether manipulation has occurred, and who the strategic actors are. In this section, we give computationally efficient methods to do so based on knowledge of the network edges and observing the corrupted equilibrium z' . The latter can be found, for example, by performing sentiment analysis on the users' posts.

Detecting Manipulation with a Hypothesis Test. In many real-world networks, the distribution of truthful opinions follows a Gaussian distribution (Figure 6). Given estimates $(\hat{L}, \hat{A})$ for the

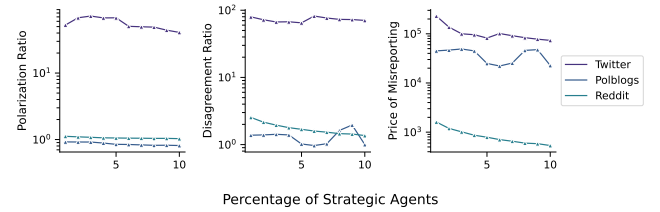


Figure 5: Polarization ratio ($\mathcal{P}(z')/\mathcal{P}(z)$), disagreement ratio ($\mathcal{D}(z')/\mathcal{D}(z)$), and price of misreporting ($C(z')/C(z)$) for the three datasets for varying the size of $|S|$. The size of $|S|$ corresponds to the top p percent of the actors ($|S| = \lceil pn \rceil$) based on their eigenvector centrality (in decreasing order), for $p \in [0.01, 0.1]$. The susceptibility parameter is set to $\alpha_i = 0.5$.

graph Laplacian and susceptibility matrix, the platform can observe the corrupted equilibrium z' and solve for the strategic opinions s' via:

$$\hat{s}' := \hat{A}^{-1}((I - \hat{A})\hat{L} + \hat{A})z'. \quad (10)$$

We propose that the platform perform a one-sample t -test with the entries s' , with a population mean $\mu_0 \in \mathbb{R}$ based on e.g. historical data. Under the null hypothesis in which no manipulation has occurred, $s' = s$, so the test should fail to reject the null hypothesis. However, when agents $S \subset [n]$ deviate, then $s'_i \neq s_i$ for $i \in S$, so the test should reject the null for large enough deviations. The test is simple, and described in Algorithm 1. Figure 6 shows the results

ALGORITHM 1: Learning from Misreporting Equilibrium with Hypothesis Testing

Input: Estimated graph information $\hat{L}$ and $\hat{A}$, observed equilibrium $\mathbf{z}'$
Output: “Manipulation” or “No Manipulation”
 Observe corrupted equilibrium $\mathbf{z}'$.
 Solve for $\hat{\mathbf{s}}'$ (Eq. (10)).
 Perform one-sample t test on the entries of $\hat{\mathbf{s}}'$ with a population mean μ_0 under the null hypothesis.
return If the t -test rejects, return “Manipulation.” Otherwise, return “No Manipulation.”

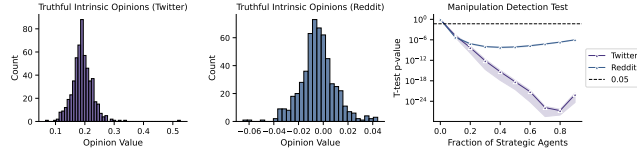


Figure 6: The true opinions for Twitter (left) and Reddit (middle) both follow a normal distribution. When simulating strategic manipulation with random choices of S (right), the detection test (Algorithm 1) has no Type I or Type II error at significance level 0.05. The tester uses $\hat{L} = L$, $\hat{A} = A = \frac{1}{2}I$, and μ_0 equal to the mean of the true intrinsic opinions. Shaded regions are 95% confidence intervals for p -values of the test across 5 independent runs.

of the test for varying choices of S . We see that at significance level 0.05, the test has low Type I error, as it will return “No Manipulation” when $S = \emptyset$, and low Type II error as it will return “Manipulation” when $S \neq \emptyset$.

For the Political Blogs dataset, intrinsic opinions belong to $\{\pm 1\}$, so the null hypothesis should be a biased Rademacher distribution. In this case, one should use a χ^2 -test, as in Agarwal et al. [3], Chen and Tsourakakis [17].

Learning the Strategic Actors with Robust Regression. The algorithm described in the previous section (Algorithm 1) can be used to detect whether there exists manipulation in the network. However, the set S is unknown, and therefore the platform cannot target the deviators to perform interventions to mitigate strategic behavior.

It is, therefore, essential for the platform to be able to identify the set of deviators S , in case the platform needs to take regulatory actions. While at first, it may seem that finding the set of deviators S is a hard task, it turns out that under mild assumptions on the intrinsic opinion formation process, we can learn the set of deviators S from observing the strategically corrupted equilibrium $\mathbf{z}'$ via Algorithm 2 in polynomial time, described in Algorithm 2. Our algorithm is based on robust regression leveraging the TORRENT algorithm developed by Bhatia et al. [7] and requires access to a node embedding matrix $X \in \mathbb{R}^{n \times d}$, and the size $|S|$ of the set of deviators.

The key idea of Algorithm 2 is that if the size of the strategic set S is sufficiently small, in general, $|S| \leq Cn$ for some small constant C , then we can view the misreported intrinsic opinions $\mathbf{s}'$ as a perturbation of the truthful opinion vector $\mathbf{s}$, and then use a robust regression algorithm to estimate $\mathbf{s}$. We assume that the embedding

ALGORITHM 2: Learning from Misreporting Equilibrium

Input: Features $X \in \mathbb{R}^{n \times d}$, graph information L and A , observed equilibrium $\mathbf{z}'$, set size $|S|$
Output: Set of strategic agents $\hat{S}$, estimated intrinsic beliefs $\hat{\mathbf{s}}$.
 $\hat{\mathbf{s}}' \leftarrow A^{-1}((I - A)L + A)\mathbf{z}'$;
 $\hat{\mathbf{v}} \leftarrow$ Robust Regression (TORRENT) with design matrix X , response vector $\hat{\mathbf{s}}'$;
 $\hat{\mathbf{s}} \leftarrow X\hat{\mathbf{v}}$;
 $\text{diffs} \leftarrow |\hat{\mathbf{s}} - \hat{\mathbf{s}}'|$;
 $\hat{S} \leftarrow$ indices of top k largest values in diffs ;
return $\hat{\mathbf{s}} \in \mathbb{R}^n$, $\hat{S} \subseteq [n]$

matrix $X \in \mathbb{R}^{n \times d}$ determines intrinsic beliefs: for example, demography, geographic location, etc. Node-level features can be learned by a variety of methods, such as spectral embeddings on the graph Laplacian or graph neural networks. Previous works have used the framework of combining a robust estimator with model-specific information to learn from “strategic sources” of data, such as in bandits [37], controls [56], and network formation games [35].

In the sequel, we give the precise technical condition of the features required for robust regression to work [7], which is based on conditions on the minimum and maximum eigenvalues of the correlation matrix determined by the features corresponding to agents in S . Specifically, for a matrix $X \in \mathbb{R}^{n \times d}$ with n samples in $\mathbb{R}^d$ and $S \subset [n]$ let $X_S \in \mathbb{R}^{|S| \times d}$ select rows in S . Note that $\lambda_{\min}(\cdot)$, $\lambda_{\max}(\cdot)$ are the min/max eigenvalues respectively.

DEFINITION 4.1 (SSC AND SSS CONDITIONS). Let $\gamma \in (0, 1)$. The features matrix $X \in \mathbb{R}^{n \times d}$ satisfies the Subset Strong Convexity Property at level $1 - \gamma$ and Subset Strong Smoothness Property at level γ with constants $\xi_{1-\gamma}$, Ξ_γ respectively if:

$$\xi_{1-\gamma} \leq \min_{S \subset [n]: |S|=(1-\gamma)n} \lambda_{\min}(X_S^T X_S),$$

$$\Xi_\gamma \geq \max_{S \subset [n]: |S|=\gamma n} \lambda_{\max}(X_S^T X_S).$$

We give our guarantee for Algorithm 2.

PROPOSITION 4.2. Let X be as in Algorithm 2, and suppose that $X\mathbf{v} = \mathbf{s}$ for some $\mathbf{v} \in \mathbb{R}^d$, and that X satisfies the SSC condition at level $1 - \gamma$ with constant $\xi_{1-\gamma}$, and SSS condition at level γ with constant Ξ_γ (Definition 4.1). Then, there exist absolute constants $C, C' > 0$ such that if $|S| \leq Cn$ and $4 \frac{\sqrt{\Xi_\gamma}}{\sqrt{\xi_{1-\gamma}}} < 1$, Algorithm 2 returns $\hat{\mathbf{s}}$ such that $\|\hat{\mathbf{s}} - \mathbf{s}\|_2 \leq \|X\|_2 n^{-\omega(1)}$, using $T = C'(\log n)^2$ iterations of TORRENT for the Robust Regression step. Moreover, if for all $j \in S$ we have $|\mathbf{s}_j - \mathbf{s}'_j| \gg \|X\|_2 n^{-\omega(1)}$, then $\hat{S} = S$.

For the proof, we first state the technical result of Bhatia et al. [7] that we require.

THEOREM 4.3 (BHATIA ET AL. [7]). Let $X \in \mathbb{R}^{n \times d}$ be a design matrix and $C > 0$ an absolute constant. Let $\mathbf{b} \in \{0, 1\}^n$ be a corruption vector with $\|\mathbf{b}\|_0 \leq \alpha n$, for $\alpha \leq C$. Let $\mathbf{y} = X\mathbf{w}^* + \mathbf{r}$ be the observed responses, and $\gamma \geq \alpha$ be the active set threshold given to the Algorithm 2 of Bhatia et al. [7]. Suppose X satisfies the SSC property at level $1 - \gamma$ and SSS property at level γ , with constants $\xi_{1-\gamma}$ and Ξ_γ

respectively. If the data $(X, \mathbf{y})$ are such that $\frac{4\sqrt{\Xi_Y}}{\sqrt{\Xi_1 - \gamma}} < 1$, then after t iterations, Algorithm 2 of Bhatia et al. [7] with active set threshold $\zeta \geq \gamma$ obtains a solution $\mathbf{w}^T \in \mathbb{R}^d$ such that, for large enough n ,

$$\|\mathbf{w}^T - \mathbf{w}^*\|_2 \leq \frac{\|\mathbf{r}\|_2}{\sqrt{n}} \exp(-cT).$$

It is interesting to investigate what an upper bound on the size of S is when nodes have community memberships, such that recovery is possible, as the SSC and SSS conditions determine it. Specifically, we show the following for a blockmodel graph (proof deferred in the Appendix).

PROPOSITION 4.4. *If G has two communities with n_1 and n_2 nodes respectively such that $n_1 \geq n_2 \geq 1$, and $X \in \{0, 1\}^{n \times 2}$ is an embedding vector where each row $\mathbf{x}_i$ corresponds to a one-hot vector for the community of node i , then Algorithm 2 can recover S perfectly as long as $|S| < \frac{n_1}{17}$.*

For instance, when Theorem 4.4 is applied to the Polblogs dataset, it shows that S can be fully recovered as long as $|S| \leq 9$. We can obtain a slightly worse bound and extend the result to a blockmodel graph with K communities (proof in the Appendix).

PROPOSITION 4.5. *If G has $K \geq 2$ communities with sizes $n_1 \geq n_2 \geq \dots \geq n_K$ with $(\frac{16K}{16K+1}) \frac{n}{K} < n_K \leq \frac{n}{K}$ and $X \in \{0, 1\}^{n \times K}$ is an embedding vector where $\mathbf{x}_i$ corresponds to an one-hot encoding of the community membership, then Algorithm 2 can recover S perfectly as long as $|S| < \frac{1}{16K+1}n$.*

In detail, Theorem 4.5 states that as long as the smallest community of the graph has size $\Theta(n/K)$ then the recovery of a set of size $|S| = O(n/K)$ is possible. If $|S| \gg n/K$ and S contains all members of the smallest community, then robust regression can fail. We show that in real-world datasets, Algorithm 2 can identify the set of deviators with high accuracy (cf. Figure 7). Specifically, in the real-world datasets we take S to be a randomly sampled set of size $\lceil pn \rceil$ for $p \in \{0.05, 0.1, 0.15, 0.2\}$ and the embeddings to be 128-dimensional Node2Vec embeddings for Twitter and Reddit and community membership embeddings for the Polblogs dataset. Algorithm 2 achieves low recovery error as well as high balanced accuracy score.

5 Managerial Insights and Discussion

5.1 Platform Governance

Our characterization of the Nash Equilibrium in multi-adversary opinion dynamics offers three strategic pillars for social media platform governance:

PoM as a Vulnerability Metric. The *Price of Misreporting* (PoM) quantifies the social cohesion lost to strategic dishonesty. Platform stakeholders should prioritize monitoring networks with high connectivity but low individual susceptibility, as these “viral” environments are most vulnerable to disproportionate increases in polarization from even small-scale coordinated misreporting.

Beyond Content Takedowns. Because strategic actors optimize reported opinions based on network structure, they can exert significant influence without triggering traditional platform health interventions (for example, flagging a post as fake news). Platforms

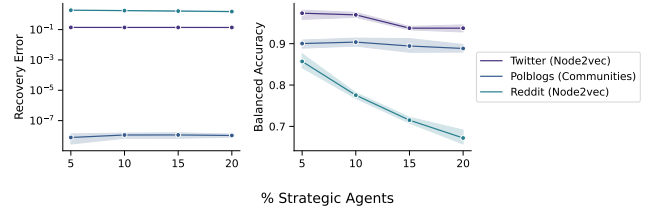


Figure 7: Reconstruction error and balanced accuracy for the robust regression problem in Algorithm 2 as a function of the fraction of strategic agents. The left panel reports the recovery error $\frac{1}{n} \sum_{i \in [n]} \left| \frac{\hat{s}_i - s_i}{s_i} \right|$, while the right panel shows the balanced accuracy between the recovered set $\hat{S}$ and the true set S . Confidence intervals are obtained by sampling S uniformly at random from $[n]$ five times for each $|S|$. For the Twitter and Reddit datasets, we use 128-dimensional Node2vec embeddings. For Polblogs, node features correspond to community membership (liberal $\mathbf{x}_i = (1, 0)$, conservative $\mathbf{x}_i = (0, 1)$), with intrinsic opinions given by $\mathbf{s} = X\mathbf{v}$ for $\mathbf{v} = (1, -1)^T$. We also report results using 128-dimensional spectral embeddings. For TORRENT, the recovery threshold is set to $|S|/n$ and the step size to $\eta = 1/\|\tilde{X}\|_2^2$, where $\tilde{X}$ denotes the min-max normalized embedding matrix.

must shift from individual post moderation to monitoring *structural shifts* and community sensitivity using “what-if” simulations of the Nash Equilibrium.

Structural Friction. To narrow the gap between reported and intrinsic opinions, platforms can effectively increase the sincerity cost by implementing friction mechanisms. Increasing transparency of historical stances and cross-network consistency makes maintaining strategic “lies” more costly, disincentivizing extreme misreporting.

5.2 Conclusion

We introduced a game-theoretic framework that replaces the “single bad actor” assumption with a realistic model of multiple, competing adversaries. We proved the existence of a unique Nash Equilibrium and introduced the Price of Misreporting to bound social inefficiency. Our empirical results across Reddit, Twitter, and Political Blogs datasets demonstrate that multi-agent competition often heightens global disagreement rather than neutralizing it. These findings provide a mathematical foundation for a new generation of intervention tools that prioritize the structural integrity of social graphs over keyword-based moderation. Future research should investigate how these strategic behaviors evolve alongside adaptive network topologies.

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A Proofs

Proof of Theorem 2.2. Consider agent $i \in S$. To calculate the best-response s'_i of i in response to s'_{-i} , we analyze derivatives of its cost function with respect to s' . Since the equilibrium z' is $z' = Bs'$, we have:

$$\begin{aligned} c_i(z') &= (1 - \alpha_i) \sum_{j \sim i} w_{ij} (z'_i - z'_j)^2 + \alpha_i (z'_i - s_i)^2 \\ c_i(s') &= (1 - \alpha_i) \sum_{j \sim i} w_{ij} ((e_i - e_j)^T Bs')^2 + \alpha_i (e_i^T (Bs' - s))^2 \\ &= (1 - \alpha_i) \sum_{j \sim i} w_{ij} (s')^T (B^T (e_i - e_j) (e_i - e_j)^T B) (s') \\ &\quad + \alpha_i ((s')^T B^T e_i e_i^T Bs' - 2(s')^T B^T e_i e_i^T s + s^T e_i e_i^T s) \\ \nabla_{s'} c_i(s') &= (1 - \alpha_i) \sum_{j \sim i} w_{ij} 2(B^T (e_i - e_j) (e_i - e_j)^T B) (s') \\ &\quad + \alpha_i (2B^T e_i e_i^T Bs' - 2B^T e_i e_i^T s), \\ \nabla_{s'}^2 c_i(s') &= 2(1 - \alpha_i) B^T \left[\sum_{j \sim i} w_{ij} (e_i - e_j) (e_i - e_j)^T \right] B + 2\alpha_i B^T e_i e_i^T B. \end{aligned}$$

Let $L_i \in \mathbb{R}^{n \times n}$ be:

$$L_i := \sum_{j \sim i} w_{ij} (e_i - e_j) (e_i - e_j)^T.$$

Notice that L_i is precisely the Laplacian of the graph when all edges not incident to i are equal to zero. Therefore $L_i \succeq 0$. Since $e_i e_i^T \succeq 0$, the Hessian of c_i with respect to s' is PSD. In particular, its (i, i) entry is non-negative, so $\frac{\partial^2 c_i(s')}{\partial (s'_i)^2} \geq 0$, and hence the optimal s'_i is at the critical point. This is given as:

$$\begin{aligned} 0 &= \frac{1}{2} \frac{\partial}{\partial s'_i} c_i(s') \\ &= e_i^T (1 - \alpha_i) B^T \left[\sum_{j \sim i} w_{ij} (e_i - e_j) (e_i - e_j)^T \right] Bs' \\ &\quad + e_i^T \alpha_i (B^T e_i e_i^T Bs' - B^T e_i e_i^T s) \\ &= (1 - \alpha_i) e_i^T B^T L_i Bs' + e_i^T \alpha_i (B^T e_i e_i^T Bs' - B^T e_i e_i^T s). \end{aligned}$$

The above display gives the solution for s'_i in terms of all entries of s' . Assembling the critical points into a linear system, we obtain precisely that for all $i \in S$, $e_i^T T_i s' = y_i$. Since $s'_j = s_j$ for $j \notin S$, the overall linear system describes the Nash equilibria.

Proof of Theorem 4.4. Let V_1 correspond to the vertex set for community 1 and V_2 correspond to the vertex set for community 2. Let I_q denote the $q \times q$ identity matrix, and $a_i = |V_i \cap S|$ for $i = 1, 2$. Then

$$X_S^T X_S = \begin{pmatrix} I_{a_1} & 0 \\ 0 & I_{a_2} \end{pmatrix}.$$

Therefore $\lambda_{\max}(X_S^T X_S) = \max\{|V_1 \cap S|, |V_2 \cap S|\}$ and $\lambda_{\min}(X_S^T X_S) = \min\{|V_1 \cap S|, |V_2 \cap S|\}$.

First, we determine sufficient ranges of γ and the value of Ξ_γ : Let S be such that $|S| = \gamma n$. We have the following options:

- $S \subseteq V_1$. Then $\lambda_{\max}(X_S^T X_S) = \gamma n$.
- $S \subseteq V_2$. Then $\lambda_{\max}(X_S^T X_S) = \gamma n$.
- $V_1 \subseteq S$. Then $\lambda_{\max}(X_S^T X_S)$ lies between $\gamma n/2$ and γn .
- $V_2 \subseteq S$. Then $\lambda_{\max}(X_S^T X_S)$ lies between $\gamma n/2$ and γn .
- If S lies partially in V_1 and V_2 , then $\lambda_{\max}(X_S^T X_S) = \max\{(1 - t)\gamma n, t\gamma n\}$ for some $t \in [0, 1]$. Again, this is upper bounded by γn .

The above yield $\Xi_\gamma = \gamma n$. To determine $\xi_{1-\gamma}$ we let S be such that $|S| = (1 - \gamma)n$. We have the following options:

- $S \subseteq V_1$. Then $\lambda_{\min}(X_S^T X_S) = 0$.
- $S \subseteq V_2$. Then $\lambda_{\min}(X_S^T X_S) = 0$.
- $V_1 \subseteq S$. Then $\lambda_{\min}(X_S^T X_S) = \min\{n_1, (1 - \gamma)n - n_1\} = n_1 \geq (1 - \gamma)n - n_2$ since always $n_1 \geq (1 - \gamma)n/2$.
- $V_2 \subseteq S$. Then $\lambda_{\min}(X_S^T X_S) = \min\{n_2, (1 - \gamma)n - n_2\} = (1 - \gamma)n - n_2$ since $n_2 \leq (1 - \gamma)n/2$.
- If S lies partially in V_1 and V_2 , then $\lambda_{\min}(X_S^T X_S) = \min\{(1 - t)(1 - \gamma)n, t(1 - \gamma)n\}$ for some $t \in [0, 1]$. Again, this is lower bounded by $(1 - \gamma)n - n_2$.

Therefore, for either $1 - \gamma \leq n_1/n$ or $1 - \gamma \leq n_2/n$ we have that $\xi_{1-\gamma} = (1 - \gamma)n - n_2$. The final inequality corresponds to

$$4\sqrt{\frac{\Xi_\gamma}{\xi_{1-\gamma}}} < 1 \iff \gamma < \frac{1}{17} - \frac{n_2}{17n}$$

Combining the above we get two systems of inequalities. The first one corresponds to $1 - \frac{n_1}{n} \leq \gamma < \frac{1}{17} - \frac{n_2}{17n}$ which holds for $n_2 < 1/18$ which is impossible since $n_2 \geq 1$. The second one corresponds to $1 - \frac{n_2}{n} \leq \gamma < \frac{1}{17} - \frac{n_2}{17n}$ which holds for $n_2 > 16/18$, which is always true.

Proof of Theorem 3.1. First, we set $\tilde{\alpha} = \alpha/(1 - \alpha)$. By substituting $z = Bs$ we can show by straightforward algebra that $C(z)/(1 - \alpha) = s^T Qs$ where $Q \succeq 0$ with

$$Q = BLB + \tilde{\alpha}(I - 2B + B^2) \quad (11)$$

Since $Q \succeq 0$, it has eigendecomposition $Q = U \Lambda_Q U^T$. Moreover, U is precisely the matrix of eigenvectors for the Laplacian. The eigenvalues of Q can be shown to be $\tilde{\alpha}^2/(\lambda_i + \tilde{\alpha})$. Therefore, $C(z) = s^T Qs \geq \frac{\tilde{\alpha}^2}{\lambda_n + \tilde{\alpha}} \|s\|_2^2$.

Next, let $\text{diag}(B)$ be the diagonal matrix with entries B_{ii} and $\overline{\text{diag}}(B)$ be the diagonal matrix with elements B_{ii} . It is easy to show that $s' = (1/\tilde{\alpha})B^{-1}\overline{\text{diag}}(B)s$ and $z' = (1/\tilde{\alpha})\overline{\text{diag}}(B)s$, which similarly implies (after algebraic operations) that $C(z')/(1 - \alpha) = s'^T Q's$ where:

$$Q' := \frac{1}{\tilde{\alpha}^2} \overline{\text{diag}(B)} L \overline{\text{diag}(B)} + \frac{1}{\tilde{\alpha}} B^{-1} \left(\overline{\text{diag}(B)} \right)^2 B^{-1} - 2 \frac{1}{\tilde{\alpha}} B^{-1} \left(\overline{\text{diag}(B)} \right)^2 + \frac{1}{\tilde{\alpha}} \left(\overline{\text{diag}(B)} \right)^2$$

Note that Q' cannot be diagonalized since, in general, $\overline{\text{diag}(B)}$ has a different eigenbasis than L . However, we note that:

$$\|\overline{\text{diag}(B)}\|_2 = \max_i B_{ii} \leq \|B\|_2 = 1, \quad (12)$$

$$\|B^{-1}\|_2 = \max_i \frac{\lambda_i + \tilde{\alpha}}{\tilde{\alpha}} = \frac{\lambda_n + \tilde{\alpha}}{\tilde{\alpha}}. \quad (13)$$

By the triangle inequality, the Cauchy-Schwarz inequality, and Equations (12) and (13), we have that:

$$\begin{aligned} \|Q'\|_2 &\leq \frac{1}{\tilde{\alpha}^2} \|L\|_2 \left(\|\overline{\text{diag}(B)}\|_2 \right)^2 + \frac{1}{\tilde{\alpha}} \left(\|\overline{\text{diag}(B)}\|_2 \right)^2 \|B^{-1}\|_2^2 \\ &\quad + \frac{2}{\tilde{\alpha}} \|B^{-1}\|_2 + \frac{1}{\tilde{\alpha}} \left(\|\overline{\text{diag}(B)}\|_2 \right)^2 \\ &\leq \frac{(\lambda_n + 4\tilde{\alpha})(\lambda_n + \tilde{\alpha})}{\tilde{\alpha}^3}. \end{aligned}$$

Therefore $C(z')/(1-\alpha) \leq \frac{(\lambda_n + 4\tilde{\alpha})(\lambda_n + \tilde{\alpha})}{\tilde{\alpha}^3} \|s\|_2^2$. Hence,

$$\frac{C(z')}{C(z)} \leq \frac{(\lambda_n + 4\tilde{\alpha})(\lambda_n + \tilde{\alpha})^2}{\tilde{\alpha}^5}. \quad (14)$$

Finally, we can simplify:

$$\frac{(\lambda_n + 4\tilde{\alpha})(\lambda_n + \tilde{\alpha})^2}{\tilde{\alpha}^5} \leq \frac{64(\lambda_n + \tilde{\alpha})^3}{\tilde{\alpha}^5} \leq \frac{128(\max\{\lambda_n, \tilde{\alpha}\})^3}{\tilde{\alpha}^5}. \quad (15)$$

Extension to heterogeneous α_i . From Theorem 3.1, we can show that the upper bound is minimized when $\lambda_n = \Theta(\tilde{\alpha}^3)$ and has a value of $O(1/\tilde{\alpha}^2)$. As we noted, Theorem 3.1 can be written with $d_{\max}$ in the place of λ_n as well.

Next, we give an easy generalization to the case of differing susceptibility.

COROLLARY A.1 (PRICE OF MISREPORTING FOR HETEROGENEOUS SUSCEPTIBILITY). *If the α_i are differing, let $\alpha_{\min} = \min_i \alpha_i$ and $\alpha_{\max} = \max_j \alpha_j$. Define $\tilde{\alpha}_{\min} = \frac{\alpha_{\min}}{1-\alpha_{\max}}$, $\tilde{\alpha}_{\max} = \frac{\alpha_{\max}}{1-\alpha_{\min}}$. The Price of Misreporting is bounded as:*

$$\text{PoM} \leq \frac{1-\alpha_{\min}}{1-\alpha_{\max}} \frac{(\lambda_n + 4\tilde{\alpha}_{\max})(\lambda_n + \tilde{\alpha}_{\max})^2}{\tilde{\alpha}_{\min}}.$$

Note that $C(z') \leq (1-\alpha_{\min})(z')^T L z' + \alpha_{\max} \|z' - s\|_2^2 = \overline{C}(z')$, and $C(z) \geq (1-\alpha_{\max})z^T L z + \alpha_{\min} \|z - s\|_2^2 = \underline{C}(z)$ where $\alpha_{\min} = \min_{i \in [n]} \alpha_i$, and $\alpha_{\max} = \max_{i \in [n]} \alpha_i$. Then, the same analysis of Theorem 3.1 can be applied, since $\overline{C}(z')/\underline{C}(z) \geq C(z')/C(z)$.

Proof of Theorem 4.2. Let $\hat{s}'$ be as in Algorithm 2, and $y = \hat{s}'$. Notice $y = s + (s' - s) + (\hat{s}' - s')$. Let $w := (s' - s)$ be the corruption vector due to strategic negotiations and $r = (\hat{s}' - s')$ be the residual vector due to least-squares regression. We claim that $r = 0$, because A^{-1} is full rank and $((I - A)L + A)$ is full rank, so $\hat{s}' = A^{-1}((I - A)L + A)z' = s'$.

Next, we apply Theorem 4.3. Notice that $\|w\|_0 \leq Cn$ by assumption. Moreover, X satisfies the SSC and SSS conditions. Therefore, after T iterations, Algorithm 2 obtains $\hat{s}$ such that:

$$\|\hat{\theta} - v\|_2 \leq \frac{\exp(-cT)}{\sqrt{n}} \|s' - s\|_2.$$

Therefore, letting $T = C'(\log n)^2$ for large enough constant $C' > 0$, we see that $\|\hat{\theta} - v\|_2 \leq O(n^{-\omega(1)})$. Hence $\|\hat{s} - s\|_2 = \|X\hat{\theta} - Xv\|_2 \leq \|X\|_2 \cdot n^{-\omega(1)}$. Now, let $u = \hat{s} - s'$. If $i \in [n] \setminus S$, then $|u_i| \leq \|X\|_2 \cdot n^{-\omega(1)}$. On the other hand for $j \in S$, $|u_j| \geq |s_j - s'_j| - \|X\|_2 n^{-\omega(1)}$. Therefore the top- $|S|$ entries of u recover S .

Proof of Theorem 4.5. First, we note that if $V_1, \dots, V_K$ are the vertex sets and S is a set of size γn the maximum eigenvalue equals to $\lambda_{\max}(X_S^T X_S) = \max_{i \in [K]} |V_i \cap S|$ and is always at most γn . So $\Xi_\gamma = \gamma n$. If S is a set of size $(1-\gamma)n$, then the minimum eigenvalue $\lambda_{\min}(X_S^T X_S) = \min_{i \in [K]} |V_i \cap S|$ is maximized when $|V_1 \cap S| = \dots = |V_K \cap S| = (1-\gamma)n/K$ which holds as long as $(1-\gamma)n/K \leq n_K$, so $\xi_{1-\gamma} = (1-\gamma)n/K$ as long as $\gamma \geq 1 - n_K/nK$. Also the other condition is

$$4\sqrt{\frac{\Xi_\gamma}{\xi_{1-\gamma}}} < 1 \iff \gamma < \frac{1}{16K+1}$$

Finally, we must have $1/(16K+1) > 1 - Kn_K/K$ which yields $n_K > n/K(16K/(16K+1))$.

B Additional Figures

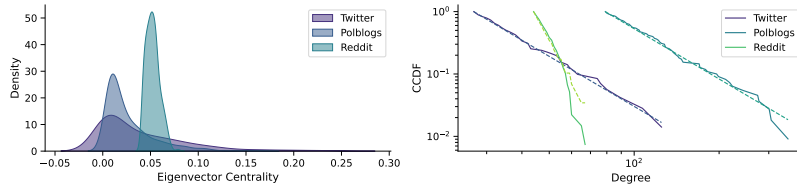


Figure 8: Distribution of centralities and degrees for the datasets

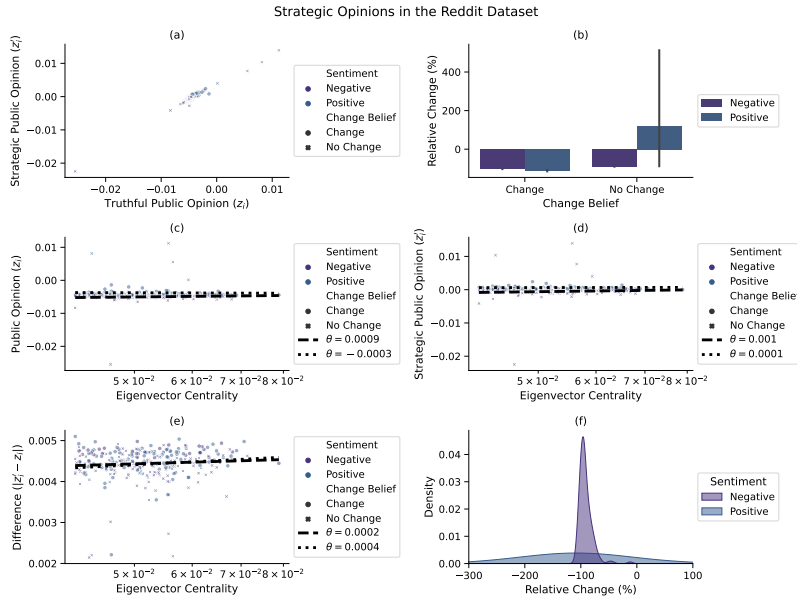


Figure 9: Running the experiments of Figure 3 for the Reddit dataset.

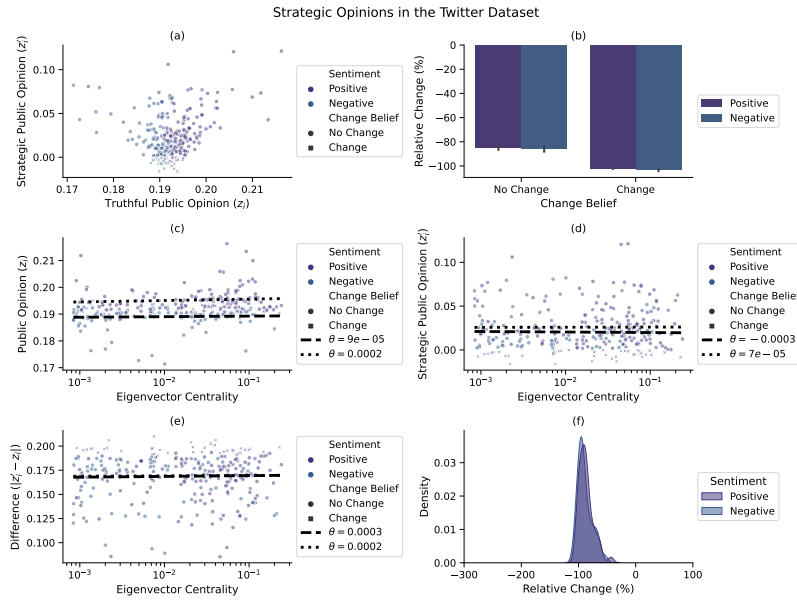


Figure 10: Running the experiments of Figure 3 for the Twitter dataset.